

Worksheet 3D: Graphing Quadratics

AIML Foundations Mathematics

Dublin and Dún Laoghaire ETB

Instructor: Josh Aaron

About This Worksheet This worksheet connects the algebra you've learned to visual understanding. You'll see how changing numbers in an equation changes the shape and position of a parabola. **GeoGebra Links:** Throughout this worksheet, you'll find links to interactive GeoGebra applets. Open these in a browser to explore the concepts hands-on.

Part A: The Basic Parabola (8 problems)

The simplest quadratic function is:

$$f(x) = x^2$$

Explore: [Basic Parabola](https://www.geogebra.org/calculator) Type $y = x^2$ into GeoGebra to see it.

1. Complete this table for $f(x) = x^2$:

x	-3	-2	-1	0	1	2	3
$f(x)$							

2. Plot these points on graph paper and connect them smoothly.
3. Answer these questions about $f(x) = x^2$:
- (a) What is the lowest point on the graph? (This is called the **vertex**)
 - (b) Is the parabola symmetric? Around what line?
 - (c) Does the parabola open upward or downward?
 - (d) What is $f(0)$? This is the **y-intercept**.
4. The graph of $f(x) = x^2$ never goes below the x-axis. What does this tell you about:
- (a) The solutions to $x^2 = 0$?
 - (b) The solutions to $x^2 = -1$?
5. Now consider $g(x) = -x^2$. Complete the table:

x	-3	-2	-1	0	1	2	3
$g(x)$							

6. How does $g(x) = -x^2$ compare to $f(x) = x^2$?
7. For $g(x) = -x^2$:
- What is the vertex?
 - Does it open upward or downward?
 - What is the maximum or minimum value?
8. **Rule:** If the coefficient of x^2 is positive, the parabola opens _____.
If negative, it opens _____.

Part B: Vertical Stretch and Compression (8 problems)

Explore: Open GeoGebra and graph these together: - $y = x^2$ (red) - $y = 2x^2$ (blue) - $y = 0.5x^2$ (green) - $y = 3x^2$ (orange)

9. Complete the table comparing $f(x) = x^2$ and $g(x) = 2x^2$:

x	-2	-1	0	1	2
$f(x) = x^2$					
$g(x) = 2x^2$					

10. How does multiplying by 2 affect the y-values?

11. Compare $f(x) = x^2$ and $h(x) = 0.5x^2$:

x	-2	-1	0	1	2
$f(x) = x^2$					
$h(x) = 0.5x^2$					

12. How does multiplying by 0.5 affect the y-values?

13. Which parabola is "narrower" (steeper): $y = x^2$ or $y = 3x^2$?

14. Which parabola is "wider" (flatter): $y = x^2$ or $y = 0.25x^2$?

15. Match each equation to its description:

Equation	Description
$y = 5x^2$	A. Same as $y = x^2$ but flipped
$y = 0.1x^2$	B. Very narrow, opens up
$y = -x^2$	C. Very wide, opens up
$y = -3x^2$	D. Narrow, opens down

16. **Rule:** The coefficient a in $y = ax^2$ affects:

- **Sign of a :
- **Determines if parabola opens _____ or _____

- ****Size of $|a|$:**** Larger means _____, smaller means _____

Part C: Vertical Shifts (8 problems)

Explore: Graph these in GeoGebra: - $y = x^2$ (red) - $y = x^2 + 3$ (blue) - $y = x^2 - 2$ (green)

17. Complete the table:

x	-2	-1	0	1	2
$f(x) = x^2$					
$g(x) = x^2 + 3$					
$h(x) = x^2 - 2$					

18. How does $y = x^2 + 3$ compare to $y = x^2$?

19. How does $y = x^2 - 2$ compare to $y = x^2$?

20. For each function, state the vertex:

(a) $y = x^2 \rightarrow$ Vertex: (____, ____)

(b) $y = x^2 + 5 \rightarrow$ Vertex: (____, ____)

(c) $y = x^2 - 4 \rightarrow$ Vertex: (____, ____)

(d) $y = x^2 + 100 \rightarrow$ Vertex: (____, ____)

21. The equation $y = x^2 + k$ shifts the parabola:

- Up by k units if k is _____
- Down by $|k|$ units if k is _____

22. Write the equation of a parabola that:

(a) Has the same shape as $y = x^2$ but vertex at $(0, 7)$

(b) Has the same shape as $y = x^2$ but vertex at $(0, -3)$

23. How many x-intercepts (roots) does each have?

(a) $y = x^2 + 1$

(b) $y = x^2$

(c) $y = x^2 - 4$

24. Connect to algebra: Solve $x^2 - 4 = 0$. How do the solutions relate to the graph of $y = x^2 - 4$?

Part D: Horizontal Shifts (8 problems)

Explore: Graph these in GeoGebra: - $y = x^2$ (red) - $y = (x - 2)^2$ (blue) - $y = (x + 3)^2$ (green)

25. Complete the table for $f(x) = (x - 2)^2$:

x	0	1	2	3	4
$(x - 2)$					
$f(x) = (x - 2)^2$					

26. What is the vertex of $y = (x - 2)^2$?

27. Complete the table for $g(x) = (x + 3)^2$:

x	-5	-4	-3	-2	-1
$(x + 3)$					
$g(x) = (x + 3)^2$					

28. What is the vertex of $y = (x + 3)^2$?

29. **The Tricky Part:**

- $y = (x - 2)^2$ shifts the parabola _____ (left/right) by 2 units
- $y = (x + 3)^2$ shifts the parabola _____ (left/right) by 3 units

The shift is OPPOSITE to the sign inside the parentheses!

30. State the vertex of each:

- (a) $y = (x - 5)^2 \rightarrow$ Vertex: (____, ____)
- (b) $y = (x + 1)^2 \rightarrow$ Vertex: (____, ____)
- (c) $y = (x - 10)^2 \rightarrow$ Vertex: (____, ____)
- (d) $y = (x + 7)^2 \rightarrow$ Vertex: (____, ____)

31. Write the equation of a parabola with:

- (a) Vertex at $(4, 0)$, same shape as $y = x^2$
- (b) Vertex at $(-2, 0)$, same shape as $y = x^2$

32. The equation $y = (x - h)^2$ has vertex at $(h, 0)$. If $h = 3$, the vertex is at _____. If $h = -4$, the vertex is at _____.

Part E: Vertex Form — Putting It All Together (10 problems)

The **vertex form** of a quadratic is:

$$y = a(x - h)^2 + k$$

Where: - (h, k) is the **vertex** - a determines **direction** (up/down) and **width**

Explore: [Vertex Form Explorer](https://www.geogebra.org/m/BUD9HAPR) *Use sliders to change a , h , and k and watch the parabola move!*

33. For $y = 2(x - 1)^2 + 3$:

- (a) What is a ? _____
- (b) What is h ? _____
- (c) What is k ? _____
- (d) What is the vertex? (____, ____)
- (e) Does it open up or down? _____
- (f) Is it narrower or wider than $y = x^2$? _____

34. For $y = -0.5(x + 2)^2 - 1$:

- (a) What is a ? _____
- (b) What is h ? _____ (*Careful with the sign!*)
- (c) What is k ? _____
- (d) What is the vertex? (____, ____)
- (e) Does it open up or down? _____
- (f) Is it narrower or wider than $y = x^2$? _____

35. Write the equation in vertex form for a parabola with:

- (a) Vertex $(3, 2)$, opens up, same width as $y = x^2$
- (b) Vertex $(-1, 4)$, opens down, same width as $y = x^2$
- (c) Vertex $(0, -5)$, opens up, twice as narrow as $y = x^2$
- (d) Vertex $(2, -3)$, opens down, half as steep as $y = x^2$

36. Match each equation to its vertex:

Equation	Vertex
$y = (x - 4)^2 + 1$	A. (4, 1)
$y = (x + 4)^2 - 1$	B. (-4, -1)
$y = (x - 4)^2 - 1$	C. (4, -1)
$y = (x + 4)^2 + 1$	D. (-4, 1)

37. Sketch (without GeoGebra) the graph of $y = (x - 2)^2 - 4$:

- (a) Vertex: _____
- (b) Opens: _____
- (c) y-intercept (plug in $x = 0$): _____
- (d) x-intercepts (set $y = 0$ and solve): _____

38. Sketch $y = -(x + 1)^2 + 9$:

- (a) Vertex: _____
- (b) Opens: _____
- (c) y-intercept: _____
- (d) x-intercepts: _____

39. A parabola has vertex (2, 5) and passes through (0, 1). Find its equation. *Hint: Start with $y = a(x - 2)^2 + 5$ and use the point to find a .*

40. A parabola has vertex (-1, -4) and passes through (1, 0). Find its equation.

41. **Maximum/Minimum Values:** For $y = a(x - h)^2 + k$: - If $a > 0$, the parabola has a _____ value of _____ at $x =$ _____
 - If $a < 0$, the parabola has a _____ value of _____ at $x =$ _____

42. A ball's height is $h(t) = -5(t - 3)^2 + 50$.

- (a) What is the maximum height?
- (b) When is the maximum height reached?
- (c) What was the initial height ($t = 0$)?

Part F: Standard Form and Converting (10 problems)

The **standard form** is:

$$y = ax^2 + bx + c$$

To find the vertex from standard form:

$$x_{\text{vertex}} = \frac{-b}{2a}$$

Then plug this x back in to find y_{vertex} .

43. For $y = x^2 - 6x + 5$:

- (a) Identify a , b , c : _____, _____, _____
- (b) Find $x_{\text{vertex}} = \frac{-b}{2a}$: _____
- (c) Find y_{vertex} by plugging in: _____
- (d) Vertex: (____, ____)

44. For $y = x^2 + 4x + 3$:

- (a) Find the vertex.
- (b) Find the y-intercept.
- (c) Find the x-intercepts by factoring.

45. For $y = -x^2 + 2x + 8$:

- (a) Find the vertex.
- (b) Does this parabola have a maximum or minimum?
- (c) Find the x-intercepts.

46. For $y = 2x^2 + 8x + 6$:

- (a) Find the vertex.
- (b) Factor to find x-intercepts.

47. **Converting Standard to Vertex Form** To convert $y = x^2 - 4x + 7$ to vertex form, complete the square: $y = x^2 - 4x + 7$ $y = (x^2 - 4x + 4) + 7 - 4 \leftarrow$ Add and subtract $(4/2)^2 = 4$ $y = (x - 2)^2 + 3$ Convert to vertex form: - (a) $y = x^2 + 6x + 10$ - (b) $y = x^2 - 2x - 3$

48. Convert to vertex form:

- (a) $y = x^2 + 8x + 20$
- (b) $y = x^2 - 10x + 21$

49. **Converting Vertex to Standard Form** Expand $y = 2(x - 3)^2 + 1$: $y = 2(x^2 - 6x + 9) + 1 = 2x^2 - 12x + 18 + 1 = 2x^2 - 12x + 19$ Convert to standard form: - (a) $y = (x + 2)^2 - 5$ - (b) $y = 3(x - 1)^2 + 4$

50. Convert to standard form:

- (a) $y = -(x - 4)^2 + 10$
- (b) $y = 2(x + 3)^2 - 7$

Part G: Roots and the Discriminant — Graphical View (10 problems)

The **x-intercepts** (roots) of a parabola are where $y = 0$.

The **discriminant** $b^2 - 4ac$ tells us how many roots to expect:

Discriminant	Roots	Graph
$b^2 - 4ac > 0$	Two distinct real roots	Crosses x-axis twice
$b^2 - 4ac = 0$	One repeated root	Touches x-axis once (vertex on x-axis)
$b^2 - 4ac < 0$	No real roots	Never crosses x-axis

Explore: [Discriminant Visualizer](https://www.geogebra.org/m/mchqmhst)

51. For each equation, calculate the discriminant and predict the number of x-intercepts: -
 - (a) $y = x^2 - 4x + 3 \rightarrow$ Discriminant: _____, x-intercepts: _____ -
 - (b) $y = x^2 - 4x + 4 \rightarrow$ Discriminant: _____, x-intercepts: _____ -
 - (c) $y = x^2 - 4x + 5 \rightarrow$ Discriminant: _____, x-intercepts: _____
52. Verify your predictions from problem 51 by:
 - (a) Factoring or using quadratic formula for each
 - (b) Sketching each parabola (they all have the same vertex x-coordinate!)
53. A parabola has equation $y = x^2 + bx + 9$. For what values of b does it:
 - (a) Cross the x-axis twice?
 - (b) Touch the x-axis exactly once?
 - (c) Never touch the x-axis?
54. The graph of $y = x^2 - 6x + k$ will touch the x-axis exactly once when $k =$ _____.
55. For $y = ax^2 + 4x + 1$:
 - (a) If $a = 1$, how many x-intercepts?
 - (b) If $a = 4$, how many x-intercepts?
 - (c) If $a = 5$, how many x-intercepts?
56. **Connecting Algebra and Geometry:** The equation $x^2 + 2x + 5 = 0$ has no real solutions. Graphically, this means the parabola $y = x^2 + 2x + 5$ _____. Find the vertex to confirm it's entirely above the x-axis.
57. Sketch a parabola that:
 - (a) Opens upward and has two x-intercepts
 - (b) Opens downward and has two x-intercepts
 - (c) Opens upward and has no x-intercepts

- (d) Opens downward and has no x-intercepts
58. A parabola has vertex $(3, -4)$ and $a = 1$.
- Write its equation.
 - Will it cross the x-axis? Why?
 - Find the x-intercepts.
59. A parabola has vertex $(2, 5)$ and $a = 1$.
- Write its equation.
 - Will it cross the x-axis? Why?
60. A parabola has vertex $(-1, 3)$ and $a = -2$.
- Write its equation.
 - Will it cross the x-axis? Why?
 - Find the x-intercepts.

Part H: Real-World Graphing Applications (6 problems)

61. A ball is thrown with height $h(t) = -5t^2 + 20t + 1$. - (a) Write in vertex form. - (b) What is the maximum height? - (c) When is maximum height reached? - (d) Sketch the graph for $t \geq 0$. - (e) When does the ball hit the ground?
62. A company's profit (in thousands) is modeled by $P(x) = -2x^2 + 24x - 54$ where x is units sold (in hundreds). - (a) Find the vertex. What does it represent? - (b) What sales level maximizes profit? - (c) What is the maximum profit? - (d) At what sales levels does the company break even ($P = 0$)?
63. A farmer has 200m of fencing for a rectangular enclosure against a barn (one side needs no fence). The area is:

$$A(w) = w(200 - 2w) = -2w^2 + 200w$$

- (a) What width maximizes the area? - (b) What is the maximum area? - (c) Sketch $A(w)$ for sensible values of w .
64. The demand for a product is $D(p) = 1000 - 25p$ where p is price in euros. Revenue is price $\times$ quantity: $R(p) = p \cdot D(p) = p(1000 - 25p) = -25p^2 + 1000p$ - (a) What price maximizes revenue? - (b) What is the maximum revenue? - (c) At what prices is revenue zero? Interpret these.
65. A suspension bridge cable forms a parabola. With the origin at the lowest point, the cable follows $y = 0.01x^2$ where x and y are in meters. - (a) How high is the cable at $x = 50$ m from center? - (b) If the towers are 100m apart, how high are they above the lowest point? - (c) Sketch the cable.

66. Water from a fountain follows $y = -0.5x^2 + 2x$ where x is horizontal distance and y is height (meters). - (a) What is the maximum height of the water? - (b) How far horizontally does the water travel? - (c) Sketch the water's path.